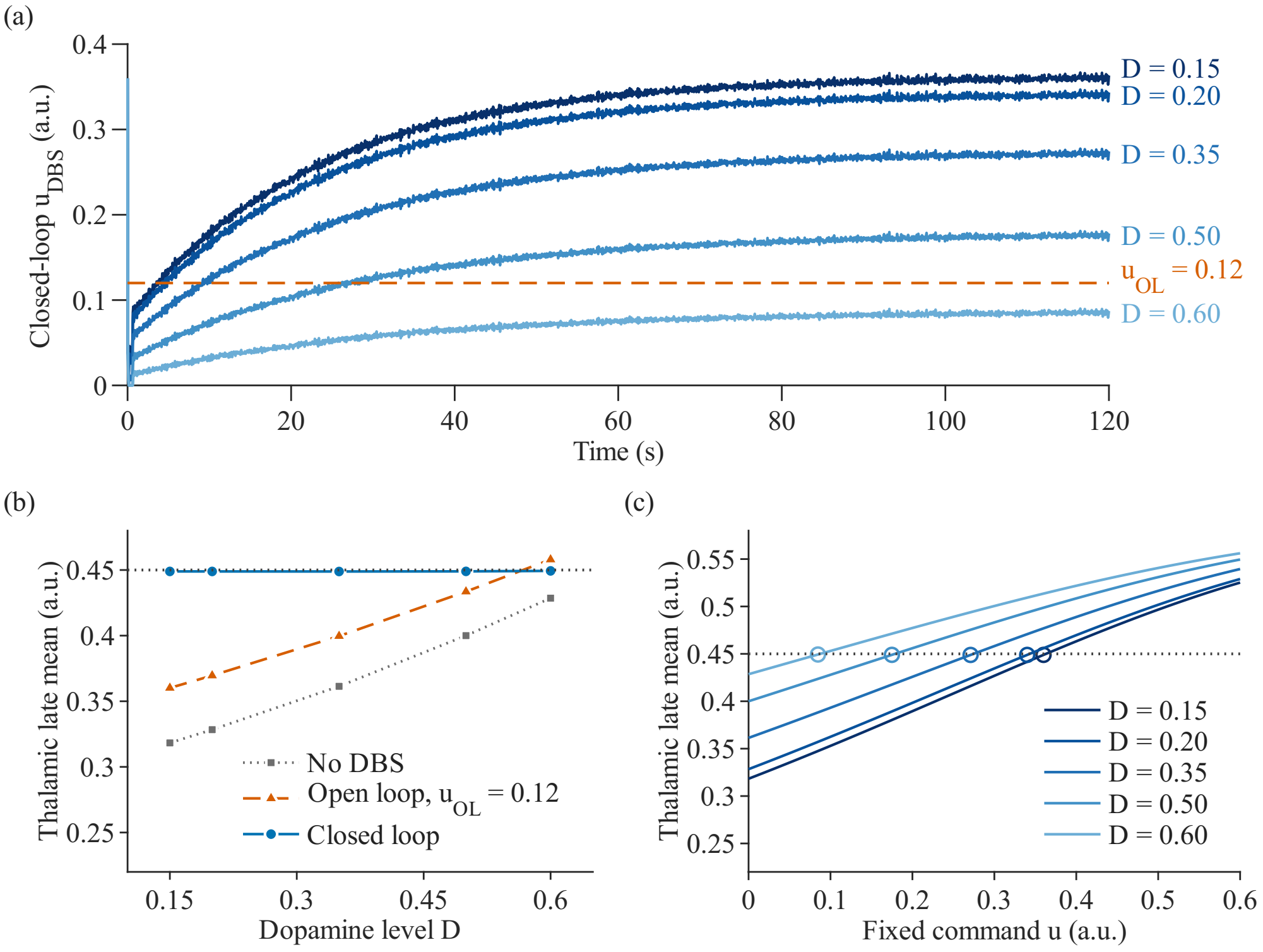

(a)
Closed-loop u_{DBS} (a.u.)
0.4
0.3
0.2
0.1
0
0
20
40
60
80
100
120
Time (s)
D = 0.15
D = 0.20
D = 0.35
D = 0.50
u_{OL} = 0.12
D = 0.60
(b)
Thalamic late mean (a.u.)
0.45
0.4
0.35
0.3
0.25
0.15
0.3
0.45
0.6
Dopamine level D
No DBS
Open loop, u_{OL} = 0.12
Closed loop
(c)
Thalamic late mean (a.u.)
0.55
0.5
0.45
0.4
0.35
0.3
0.25
0
0.1
0.2
0.3
0.4
0.5
0.6
Fixed command u (a.u.)
D = 0.15
D = 0.20
D = 0.35
D = 0.50
D = 0.60